

# WANG ZHAODONG

Design Researcher  
Master's Candidate in Design

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Design researcher investigating and reimagining social relations in shared environments through discursive design, embodied inquiry and technological mediation.

## RESEARCH INTERESTS

Public Space and Social Relations;  
Discursive and Speculative Design;  
Embodied Experience and Meaning-Making;  
Human-Technology Relations and Design Methodology

## RESEARCH CAPABILITIES

First-Person / Qualitative Inquiry  
Conceptual & Relational Modelling  
Empirical-to-Design Translation  
Metaphor/Schema Analysis • XR Prototyping

## PUBLICATIONS & MANUSCRIPTS

**Z. Wang**, & F. You (2026). From Subtle Embodied Social Experience to Designable Relations: An Image-Schema-Mediated Empirical-to-Design Method

**First author** • Under review at *ACM CHI 2027* • Preprint available on SSRN  
doi:10.2139/ssrn.7201181

F. You, X. Yan, **Z. Wang**, & Q. Fu (2026). Seeing the whole process: How cooperative process transparency in AR-HMI affects driver cognition in V2V lane changes.

**Corresponding author** • *Transportation Research Part F: Traffic Psychology and Behaviour*, 123, 103830.  
doi:10.1016/j.trf.2026.103830

Y. Chen, F. Jia, **Z. Wang**, J. Zhang, F. You, & J. Wang (2026). Designing multimodal takeover request interfaces: The impact of warning modalities and information types on drivers' takeover performance and subjective perception.

Co-author • Under review at *Transportation Research Part F: Traffic Psychology and Behaviour*

Y. Chen, Y. Jin, F. Jia, J. Wang, **Z. Wang**, B. Mehler, & F. You (2026). Beyond attention reminders: Effects of monitoring request design type and non-driving-related task engagement on driver takeover performance and subjective perceptions.

Co-author • Under review at *Accident Analysis & Prevention*

Y. Chen, J. Wang, F. Jia, X. Wu, Q. Xiao, **Z. Wang**, & F. You (2025). Is control necessary for drivers? Exploring the influence of human-machine collaboration modes on driving behavior and subjective perception under different hazard visibility scenarios.

Co-author • *Accident Analysis & Prevention*, 217, 108067. doi:10.1016/j.aap.2025.108067

## EDUCATION

9/2024 - 6/2027 **M.A. in Design** **Tongji University, Shanghai, China**  
Working thesis title: *Mixed-Reality Interaction Design for Public-Space Social Experience*  
GPA: 4.76/5.0 • Supervisor: Professor Fang You, FRSA  
Selected coursework: User Research 98/100; Narrative Studies 90/100; Service Design 96/100.

9/2020 - 6/2024 **B.Sc. in Physics** **Jilin University, Changchun, China**  
Selected coursework: Computational Physics A 96/100; Mechanics 93/100; C Programming 92/100.

## SELECTED RESEARCH PROJECTS

2025–Present **XR Design for Subtle Embodied Social Experience** **Master's Thesis Researcher, Tongji University**

- **Social and design proposition:** Used gaze as a case of subtle embodied experience to reimagine relations among strangers in public space, proposing XR imagery as an interaction mediator and discursive medium for alternative forms of public sociality.
- **Experiential modelling:** To determine what such imagery should represent, used conceptual metaphor and image-schema theories to model how gaze enters experience, acquires embodied/metaphorical meaning and informs action; distinguished substrate-level from meaning-construal affordances.
- **First-person inquiry:** To ground this model in lived experience, combined episode-focused first-person inquiry with a dual-track analysis of fleeting gaze episodes, separating problem-defining perceptual conditions from relation-providing conceptual-metaphorical relations.
- **Empirical-to-design translation:** To make open-ended experiential material operable in design, developed an image-schema-mediated method structuring it as a constrained relational design space, preserving empirical grounding while allowing designer judgment and multiple defensible forms.
- **Design case and implementation:** Applied the process with 16 participants (782 interview minutes; 160,162 Chinese transcript characters), deriving a relational grammar and three XR concepts; selected Membrane-Bubble and currently implementing it in Unity XR for laboratory evaluation.

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|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| 2025 - Present | <b>Cooperative Process Transparency in AR-HMI</b><br>Examines how different levels of cooperative process transparency affect driver cognition during Vehicle-to-Vehicle (V2V) cooperative lane changes. <ul style="list-style-type: none"><li>Contributed to the conceptualization and three-level operationalization of cooperative process transparency; curated and formally analysed performance and subjective data.</li><li>Produced research visualizations and led manuscript preparation and revision.</li></ul> | Co-Researcher, Tongji University |
| 2024 - 2025    | <b>AR Social-Space Design through Interpersonal Distance and Social Cues</b><br>Examines how virtual-agent layout, gesture amplitude, and interaction guidance shape affect, social presence, and continued engagement in AR. <ul style="list-style-type: none"><li>Contributed to research conceptualization and curated and formally analysed the experimental data.</li><li>Drafted the original manuscript and contributed to its review and revision.</li></ul>                                                       | Co-Researcher, Tongji University |
| 2024 - Present | <b>Automated-Driving HMI and Driver Attention</b><br>Comprises three driving-simulator studies of collaboration modes, monitoring requests, and multimodal takeover warnings. <ul style="list-style-type: none"><li>Contributed to the design and execution of one controlled driving-simulator study; processed and analysed eye-tracking and fNIRS data across the project studies.</li><li>Contributed to the interpretation of results and manuscript review and revision.</li></ul>                                   | Co-Researcher, Tongji University |

SELECTED DESIGN & SPATIAL PRACTICE

|              |                                                                                                                                                                                                                                                                         |                                        |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| 2023–Present | <b>ExpoCraft – Reconstruction of Expo 2010 Shanghai</b><br>Reconstruct Expo 2010 Shanghai in Minecraft from maps, photographs and archival records; translate fragmented sources into a navigable spatial archive.                                                      | Independent Designer • Ongoing Project |
| 2024         | <b>The Field of Seeds – Physical-Computing Installation</b><br>Developed the concept, analysed the site and designed interactions/circuitry; co-built a wind/proximity-responsive Arduino prototype inspired by the UK Pavilion (Seed Cathedral) at Expo 2010 Shanghai. | Co-Designer • Course Project           |
| 2024         | <b>Rooftop Garden – Site-Specific Interactive Projection</b><br>Designed a projection proposal for a typhoon-displaced roof: passersby’s silhouettes trigger projected flowers, linking interaction with architectural memory.                                          | Independent Designer • Course Proposal |
| 2023         | <b>Minetrarchi – Minecraft-Based Architectural Learning Game</b><br>Designed the concept, art, game mechanics and interactions for a Minecraft-based game concept teaching traditional Chinese timber architecture through guided tasks and free building.              | Co-Designer • Design Concept           |

EXTERNAL RESEARCH COLLABORATIONS

|           |                                                                                                                                                                                                                                    |                                                   |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| 2025      | <b>Eye-Tracking Experiment and Analysis Coordinator</b><br>Provided methodological training and supervised eye-tracking experiments and data analysis for an industry collaboration on task complexity in cloud change operations. | Tongji University & Huawei Technologies Co., Ltd. |
| 2025–2026 | <b>EEG Data Analysis Coordinator</b><br>Supervised and conducted EEG data analysis for the project “Cognitive and Neural Mechanisms Underlying the Detection and Dissemination of AI-Generated Misinformation.”                    | Beijing Institute of Petrochemical Technology     |

LEADERSHIP & CULTURAL ENGAGEMENT

|           |                                                                                                                                                                                                                                                                     |                                                               |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| 2020–2023 | <b>Photojournalist; Co-Deputy Head &amp; Head of Photography</b><br>Led photographic reporting for campus news, cultural events and public communication.                                                                                                           | Jilin University Press Corps                                  |
| 2020–2024 | <b>Deputy Head</b><br>Led college online communications and media presence; co-organised campus activities.                                                                                                                                                         | New Media Department, Physics Student Union, Jilin University |
| 2020–2022 | <b>Changchun Architectural Heritage Documentation</b><br>Studied Changchun’s Japanese-colonial (Manchukuo-era) architecture through public history writing and field photography; advocated preservation. Individual articles reached 10,000+ reads and 100+ saves. | Independent Researcher / Writer / Photographer                |

TECHNICAL SKILLS

|                      |                                                                   |
|----------------------|-------------------------------------------------------------------|
| Experimental Methods | Driving Simulation • Eye Tracking • EEG/fNIRS Analysis            |
| Research • Analysis  | SPSS • MATLAB • GraphPad Prism • NVivo • Tobii Pro Lab            |
| Programming • XR     | Unity • Python • C/C++                                            |
| Design • Media       | Figma • Adobe Photoshop • Adobe Premiere Pro • Blender • SketchUp |
| Authoring            | LaTeX                                                             |

LANGUAGES

Chinese - Native, English - Academic working proficiency, Japanese - Intermediate, French - Elementary